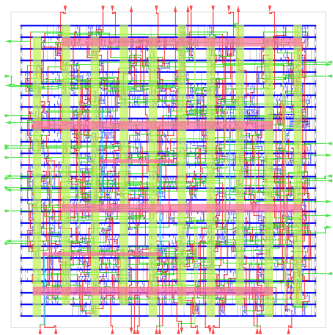
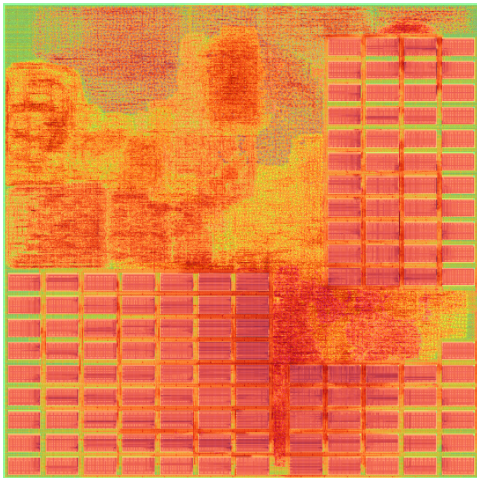


Fill, reconnect, and validate rows



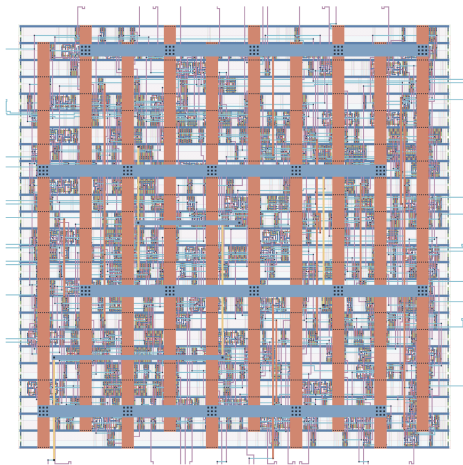
```
# Fill row gaps, reconnect global VDD/VSS,  
and check that  
# filler insertion did not create placement  
violations.  
filler_placement {FILLCELL*}  
global_connect  
check_placement -verbose
```

Power-grid checks and IR-drop data



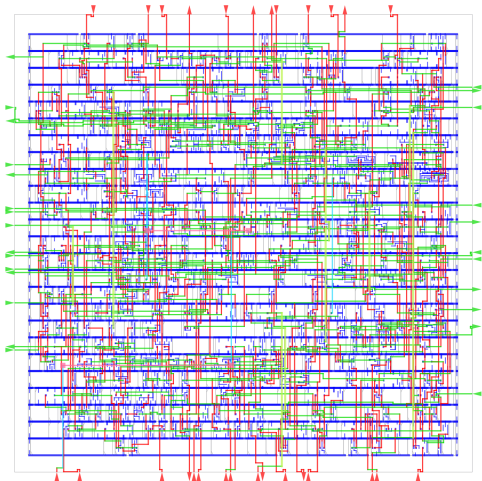
```
# Check VDD/VSS connectivity.  
check_power_grid alone does not  
# populate IR Drop. The optional solver  
writes voltage/error data.  
check_power_grid -net VDD  
check_power_grid -net VSS  
analyze_power_grid -net VDD \\  
    -source_type FULL \\  
    -voltage_file results/ariane/vdd.voltage.  
rpt \\  
    -error_file results/ariane/vdd.errors.rpt
```

Extract and reload post-route RC



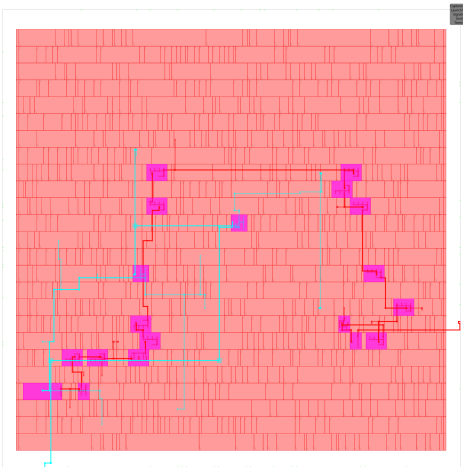
```
# Name the extraction corner, extract routed
geometry, then
# write and reload SPEF so final timing uses
post-route RC.
define_process_corner -ext_model_index 0 X
extract_parasitics -ext_model_file \\
    $openroad_root/test/Nangate45/Nangate45.
rcx_rules
write_spef results/gcd/gcd.spef
read_spef results/gcd/gcd.spef
```

Write the final hand-off files



```
# Save the final database and portable views
for downstream tools.
write_db results/gcd/gcd_final.odb
write_def results/gcd/gcd_final.def
write_verilog -remove_cells {FILLCELL*} \\
    results/gcd/gcd_final.v
write_abstract_lef results/gcd/gcd_abstract.
lef
write_sdc results/ariane/final.sdc
```

Final timing, power, and area reports



```
# Report full paths, worst slack, total  
negative slack,  
# clock skew, power, and final physical area.  
report_checks -path_delay min_max \\  
  -format full_clock_expanded \\  
  -fields {input_pin slew capacitance}  
-digits 3  
report_worst_slack -min -digits 3  
report_worst_slack -max -digits 3  
report_tns -digits 3  
report_clock_skew -digits 3  
report_power  
report_design_area
```